

Negative Exponents with Variable BasesQuestion Bank 3.T2 **ANSWER KEY**

Show your work. Circle your final answer.

For questions 1 through 6, write an equivalent expression using only positive exponents. Assume no variable equals 0.

1. a^{-6}

$$\frac{1}{a^6}$$

2. $\frac{1}{k^{-3}}$

$$k^3$$

3. $7y^{-4}$

$$\frac{7}{y^4}$$

The 7 has no negative exponent; it stays in the numerator.

4. $x^{-2}y^5$

$$\frac{y^5}{x^2}$$

5. $\frac{m^4}{n^{-3}}$

$$m^4n^3$$

6. $\frac{2p^{-5}q}{r^{-2}}$

$$\frac{2qr^2}{p^5}$$

p^{-5} moves down; r^{-2} moves up; 2 and q stay.

For questions 7 and 8, apply the laws of exponents. Write each answer using only positive exponents.

7. Write an equivalent expression.

a. $x^3 \cdot x^{-8}$

$$x^{-5} = \frac{1}{x^5}$$

b. $(3y^{-2})^3$

$$27y^{-6} = \frac{27}{y^6}$$

c. $(2a^{-3}b^2)(5a^7b^{-6})$

$$10a^4b^{-4} = \frac{10a^4}{b^4}$$

8. Write an equivalent expression.

a. $\frac{m^2}{m^9}$

$$m^{-7} = \frac{1}{m^7}$$

b. $\frac{12x^3y^{-2}}{4x^5y}$

$$3x^{-2}y^{-3} = \frac{3}{x^2y^3}$$

$$3 - 5 = -2; -2 - 1 = -3.$$

c. $\left(\frac{a^3}{4b^{-7}}\right)^5$

$$\frac{a^{15}}{4^5b^{-35}} = \frac{a^{15}b^{35}}{1,024}$$

Working backwards.

9. Find the value of n that makes each statement true.

a. $x^{-5} \cdot x^n = x^2$

$$n = 7$$

b. $\frac{y^n}{y^4} = \frac{1}{y^6}$

$$n = -2$$

$$n - 4 = -6.$$

c. $(m^n)^3 = \frac{1}{m^{12}}$

$$n = -4$$

$$3n = -12.$$

10. Let $x = 3$.

a. Find the value of x^2 and the value of x^{-2} .

9 and 1/9

b. Find the value of $-x^2$. Is it equal to x^{-2} ? Explain.

-9. No — x^{-2} is the reciprocal of x^2 , which is 1/9; $-x^2$ is the opposite of x^2 .

11. Which expression is equivalent to $5x^{-3}$?

☒ A. $\frac{5}{x^3}$

☐ B. $\frac{1}{5x^3}$

☐ C. $-5x^3$

☐ D. $\frac{1}{125x^3}$

A

12. Select ALL of the expressions that are equivalent to $\frac{1}{x^6}$.

☒ A. x^{-6}

☒ B. $x^2 \cdot x^{-8}$

☒ C. $\frac{x^3}{x^9}$

☒ D. $(x^{-2})^3$

☐ E. $-x^6$

☐ F. $(x^{-3})^{-2}$

A, B, C and D